2026 International Conference on Integrated Intelligence and Cognitive Engineering (ICIICE) | 979-8-3315-4531-4/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICIICE69672.2026.11565414

Enterprise Search Portals: Hybrid RAG at Scale for Compliance-Sensitive Knowledge Work

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***Abstract*—Enterprise search portals are starting to use large language models to produce direct answers, but regulated advisor workflows need stricter controls than normal chat systems. The system must be fast, must not leak restricted content, and must produce outputs that can be audited. This paper presents a policy-first Hybrid Retrieval-Augmented Generation (RAG) architecture for enterprise portals at scale. The design combines lexical retrieval (BM25) and dense vector retrieval, enforces access control before any text reaches reranking or generation, fuses rankings using Reciprocal Rank Fusion, and generates answers only from an evidence pack with required citations. We describe the end-to-end architecture, threat model, and security controls for prompt injection and retriever poisoning, along with an evaluation scorecard that reports retrieval quality, answer grounding, hallucination rate, audit coverage, and latency. Experiments on compliance-focused enterprise QA benchmarks show that the proposed approach improves Recall@K and answer accuracy while reducing hallucinations and achieving full audit trace coverage.**

***Index Terms*—enterprise search, retrieval-augmented generation, hybrid retrieval, compliance, audit logging, vector databases**

I. Introduction

Enterprise search portals are the main entry point to internal knowledge for many regulated teams. In advisor workflows, a single query can touch policy text, product disclosures, research notes, and client-facing templates. The portal must be fast, but speed is not enough. It must also enforce access rules, prefer the right version of content, and leave an audit trail that can be reviewed later. Traditional enterprise search returns a list of links. This works when users know what to open and how to combine sources. In practice, advisor questions are often long and mixed. A query may include a client context, a product name, a region, and a request to draft text that is compliant. Large language models can produce a direct answer, but a pure generation approach is unsafe in regulated settings. It can produce text that is not supported by policy, or it can mix content across regions and time.

Retrieval-Augmented Generation (RAG) addresses this by tying generation to retrieved evidence. A retriever selects relevant passages and the generator produces an answer conditioned on those passages [1], [2]. This improves factual support and makes it possible to show sources. However, enterprise RAG is not a simple add-on to search. In production portals, the system must handle multi-tenant isolation, role-based and attribute-based access control, strict data labels, and frequent policy updates. If access control is applied too late, restricted text may leak into the model prompt. If freshness is not handled, the system may cite an older policy that is still semantically close but no longer valid.

Retrieval quality is also harder than it looks. Lexical ranking such as BM25 is strong for exact terms, codes, and section references [3]. Dense retrieval uses embeddings to match meaning and paraphrase, and it improves recall on natural questions [4]. In enterprise portals, both are needed. Hybrid retrieval combines lexical and dense candidates and then fuses ranked lists to reduce failure cases where one signal is weak. Reciprocal Rank Fusion (RRF) is a common and effective fusion method that is simple to deploy and tune [5].

Scaling hybrid RAG adds another layer of constraints. Dense retrieval at portal scale relies on approximate nearest neighbor search to keep latency low. HNSW is a widely used graph index with strong speed and accuracy trade-offs [6]. Faiss provides a mature toolkit for building and benchmarking vector search indices and compression methods, which is useful for large, frequently updated corpora [7]. A practical portal must also manage indexing lag, cache hot queries, and keep p95 latency stable under burst traffic.

Security risks in RAG are now a first-class engineering concern. Prompt injection is a top risk because retrieved text can contain instructions that try to override system rules [8]. In addition, dense retrievers can be attacked through poisoning and backdooring so that certain queries pull attacker-chosen passages [9]. Recent work also shows coordinated prompt and corpus attacks that combine retrieval manipulation with query triggers [10]. For regulated portals, these risks are not theoretical. They directly map to data leakage, incorrect guidance, and audit failures. Risk management guidance such as the NIST AI RMF supports treating safety, transparency, and governance as part of the system design, not as an afterthought [11].

This paper targets regulated enterprise advisor portals and proposes a hybrid RAG design that is built around one principle. No model should see text that the user is not allowed to access, and no answer should be produced without traceable evidence. We focus on the engineering details that make this work at scale.

Our research contributions are as follows:

1) We define a policy-first hybrid retrieval pipeline that

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fuses BM25-style lexical retrieval with dense vector retrieval, while enforcing access control before reranking or generation [3], [4].
2) We present a rank fusion and evidence selection strategy based on RRF, with practical controls for duplication, freshness, and source trust [5].
3) We describe a scalable indexing and query plan using ANN structures and vector search toolchains suitable for large enterprise corpora [6], [7].
4) We provide a threat model for enterprise RAG and link concrete mitigations to known attack classes such as prompt injection and retriever poisoning [8], [9], [10].
5) We propose an evaluation scorecard that measures search quality, grounding, leakage risk, and operational stability in one view, aligned with risk-oriented AI guidance [11].

Section I introduces the problem and motivates Hybrid RAG for regulated enterprise portals. Section II reviews prior work in retrieval, hybrid ranking, RAG, security risks, and evaluation. Section III presents the system architecture and key components for policy-first execution and audit logging. Section IV describes the proposed methodology, including hybrid retrieval, fusion, evidence packing, and evidence-bound generation. Section V reports experimental results with tables and graphs. Section VI discusses the main findings, limitations, and operational considerations. Section VII concludes the paper and outlines future work.

II. Related Work

A. Enterprise Search Pipelines and Ranking

Enterprise search in large firms is usually built as a multi-stage ranking pipeline. A first-stage retriever must be fast and recall-heavy, and it is still common to start from lexical matching such as BM25 because it is stable, explainable, and cheap at query time [3]. However, enterprise queries are often short, full of names, and depend on context (role, product line, region). This pushes systems toward hybrid candidate generation and learned reranking.

Neural rerankers based on Transformer encoders improved passage ranking quality, but they are expensive and are typically used only after a smaller candidate set is formed [12]. As a result, most high-traffic portals use a three-step pattern: (i) retrieve many candidates, (ii) rerank top candidates, (iii) apply business and compliance rules (access filters, blocklists, and provenance constraints).

B. Dense, Sparse-Neural, and Late-Interaction Retrieval

Dense passage retrieval (DPR-style bi-encoders) made it practical to retrieve semantically related text that lexical systems miss [4]. Later work showed that even without labels, contrastive training can produce strong dense retrievers that transfer across tasks and domains [13]. For enterprise portals, these dense models help with synonymy, abbreviations, and paraphrases that are common in internal docs.

At the same time, neural sparse methods try to keep the best parts of lexical matching (token-level hits) while learning better term weighting and expansion. SPLADE is a key example and is often used as a drop-in first-stage retriever that stays compatible with inverted-index style execution [14]. Another line is late-interaction retrieval, where token-level interactions are kept but computed efficiently. ColBERT is widely used because it can outperform plain bi-encoders while still being feasible for large corpora [15].

C. Hybrid Retrieval and Fusion

Hybrid retrieval combines lexical and semantic candidates to improve recall and reduce brittle failure cases. A simple and reliable fusion method is Reciprocal Rank Fusion (RRF), which is still used because it is easy to implement and often competitive with more complex fusion rules [5]. In production systems, hybrid retrieval is commonly followed by a learned reranker, with guardrails (policy filters, citations, and audit logs) applied before answer delivery.

D. Retrieval-Augmented Generation and Grounded Answers

Retrieval-Augmented Generation (RAG) couples a retriever with a generator so the model can answer using fetched evidence rather than only its internal weights [1]. This design helps with freshness and domain specificity, both of which matter in regulated advisor portals where policies and product terms change. A known issue is that even with retrieval, the generator can still produce unsupported statements. To reduce this, newer work studies generation with explicit citations and attribution, where the output is tied to retrieved passages and the model is trained or prompted to quote sources [16], [17]. These ideas align well with compliance needs because they support traceability and review workflows.

E. Scaling Vector Search and Indexing

At enterprise scale, retrieval latency and update cost are often the main limits. Approximate nearest neighbor (ANN) indexes such as HNSW are popular due to strong recall-latency tradeoffs and good behavior in high dimensions [6]. Large-scale similarity search is also accelerated with GPU-based systems and libraries, including the FAISS ecosystem [18], [7]. In practice, systems shard indexes by business unit or sensitivity tier, and keep separate indexes for public, internal, and restricted collections to simplify access control and audit.

F. Security, Compliance Controls, and Governance for RAG Portals

RAG increases the attack surface because the model can be influenced by retrieved text. Indirect prompt injection shows that malicious instructions embedded in content can hijack downstream model behavior when that content is retrieved and appended to the prompt [19]. More recent work studies poisoning and coordinated attacks that combine prompt triggers with injected documents in the knowledge store [9], [10]. These findings motivate enterprise defenses such as content sanitization, tool-usage constraints, retrieval allowlists, and strong provenance checks, along with security testing guidance such as OWASP's LLM Top 10 [20].

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Regulated portals also require strict authorization. Attribute-based access control guidance (ABAC) and zero-trust architecture principles are often used to justify policy-first filtering and continuous verification [21], [22]. At the governance layer, risk management guidance (for example, NIST AI RMF) provides a structure for documenting model risks, controls, and monitoring plans [23].

G. Benchmarks and Evaluation for Retrieval-Augmented Systems

Benchmarking RAG needs both retrieval and generation evaluation. For retrieval quality, BEIR provides a broad set of tasks to test domain transfer, while MS MARCO remains a common supervised training and evaluation set [24], [25]. For knowledge-intensive tasks that combine retrieval and downstream language tasks, KILT provides an evaluation suite with evidence requirements [26]. For end-to-end RAG experiments, toolkits and benchmarks such as BERGEN and RAGBench help standardize evaluation across retrievers, rerankers, and generators [27], [28].

III. System Architecture

Fig. 1 shows the end-to-end portal design. The architecture separates (i) the content ingestion path and (ii) the query answering path, with a policy-first gateway and full audit traces.

A. Content Ingestion and Index Build

Enterprise content arrives from wikis, PDFs, ticket systems, and policy repositories. Each document is normalized into: *text*, *metadata*, *labels*, and *access rules*. A content hash and version ID are stored to support traceability and rollback. Documents are chunked into passages for retrieval. For each passage, the system writes: (i) a lexical posting for BM25 search [3], (ii) an embedding vector for dense search [4], and (iii) a metadata/ACL record. Vector indexing uses ANN structures (e.g., HNSW) to keep latency stable at scale [6].

B. Query Gateway (Policy-First)

All queries pass through a gateway that enforces tenant routing and authorization before retrieval. This is critical in regulated portals because it prevents restricted text from entering any reranker or generator prompt. The gateway also adds policy context (region, product scope, disclosure rules) from a policy engine.

C. Hybrid Retrieval and Fusion

The portal runs two retrieval paths in parallel: lexical top-K_1 (BM25) and dense top-K_2 (ANN vector search). Candidate sets are merged and filtered again using ACL and label rules. The final ranking is produced by Reciprocal Rank Fusion (RRF) with deduplication and freshness-aware tie breaking [5].

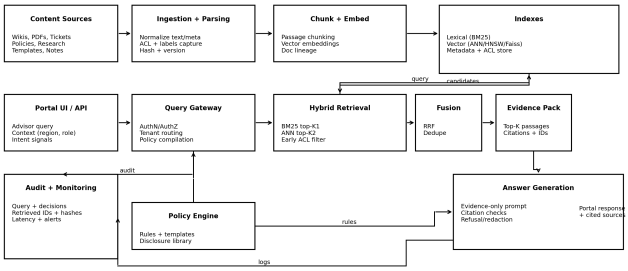


Fig. 1. Hybrid RAG portal architecture with policy-first retrieval, evidence packing, cited generation, and audit logging.

D. Evidence Pack and Answer Generation

The system builds an evidence pack containing top-K passages with stable IDs and citations. The generator receives only this evidence (plus system rules) and must produce a cited answer. If evidence is missing or conflicts, the model must refuse or ask for a narrower query. This reduces hallucination risk and supports review.

E. Audit and Monitoring

Each request writes an append-only audit record: user role, policy decisions, retrieved passage IDs, prompt hash, model version, latency, and output hash. This supports incident response and compliance review. Prompt injection is treated as a primary threat, so retrieved text is handled as untrusted input and kept separate from system instructions [20].

IV. Proposed Methodology

We propose a policy-first Hybrid RAG pipeline that produces answers only from authorized evidence and logs every step for audit.

A. Document Processing

Each document is normalized into text, metadata, labels, and access rules. It is chunked into passages (fixed token window with overlap). For every passage p, we store: (i) lexical fields for BM25 search, (ii) an embedding vector v_p for dense retrieval, (iii) ACL/label metadata inherited from the parent document.

B. Policy-First Query Handling

For each query q from user u with context c (tenant, role, region, channel), the gateway compiles an enforceable filter $F(u, c)$ that encodes RBAC/ABAC constraints and label rules. Only content that passes $F(u, c)$ is eligible for retrieval and generation.

C. Hybrid Retrieval and Fusion

We run two retrieval paths in parallel on the allowed corpus:

$$C_{\text{lex}} \leftarrow \text{Top}_{K_1}(\text{BM25}(q)), \quad C_{\text{dense}} \leftarrow \text{Top}_{K_2}(\text{ANN}(e(q)))$$

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where $e(\cdot)$ is the query encoder. We union candidates and re-check filters:

$$C \leftarrow \{d \in (C_{\text{lex}} \cup C_{\text{dense}}) \mid F(u, c, d) = 1\}.$$

We rank C using Reciprocal Rank Fusion (RRF) with deduplication and freshness-aware tie breaking, and select the top-K passages as evidence E.

D. Evidence-Bound Answer Generation

We build an evidence pack from E containing passage IDs, excerpts, and source links. The generator receives only (q, rules, E) and must: (i) produce an answer whose key claims cite passages in E, (ii) refuse when evidence is insufficient or conflicting, (iii) apply redaction rules for sensitive tokens.

E. Audit Trace

For each request, we persist an append-only trace: user/role, compiled policy filter hash, retrieved passage IDs, ranking scores, prompt hash, model version, output hash, and latency. This enables post-hoc review and compliance verification.

The proposed work uses a policy-first Hybrid RAG pipeline for enterprise search, where access control is applied before any retrieval, reranking, or answer generation so that restricted text never reaches the model. As shown in Algorithm 1, the method first compiles a policy filter from the user role, tenant, region, and request context, and then limits the searchable corpus to only authorized content. After that, it runs lexical retrieval with BM25 and semantic retrieval with dense embeddings in parallel, merges both candidate sets, applies policy checks again, and removes duplicate passages. The remaining passages are ranked with Reciprocal Rank Fusion, while freshness is used to break ties so that newer valid content is preferred. The top ranked passages are then packed as evidence with stable passage identifiers and source links. The generator receives only this evidence pack and must produce a cited answer from these passages; if the evidence is weak, missing, or conflicting, the system does not guess and instead returns a refusal or asks for a narrower query. Finally, the full request flow is stored as an audit trace with user context, policy decision, retrieved passage IDs, prompt hash, output hash, and latency, which makes the whole pipeline safe, traceable, and suitable for compliance-sensitive enterprise use.

Algorithm 1: Policy-First Hybrid RAG for Compliance-Sensitive Enterprise Search

Input: User query q, user context u, policy context c, document corpus $\mathcal{D}$,
lexical retriever $\text{BM25}(\cdot)$, dense encoder $e(\cdot)$, ANN index,
fusion constant k_{rrf}, evidence size K
Output: Cited answer a with audit trace $\mathcal{T}$

/* (1) policy-first filtering before retrieval */
$F(u, c) \leftarrow \text{CompilePolicyFilter}(u, c)$
$\mathcal{D}_{\text{allow}} \leftarrow \{d \in \mathcal{D} \mid F(u, c, d) = 1\}$
/* (2) run lexical and dense retrieval on allowed corpus */
$C_{\text{lex}} \leftarrow \text{TopK}_1(\text{BM25}(q, \mathcal{D}_{\text{allow}}))$
$\mathbf{z}_q \leftarrow e(q)$
$C_{\text{dense}} \leftarrow \text{TopK}_2(\text{ANN}(\mathbf{z}_q, \mathcal{D}_{\text{allow}}))$
/* (3) merge, re-check policy, and remove duplicates */
$C \leftarrow C_{\text{lex}} \cup C_{\text{dense}}$
$C' \leftarrow \{d \in C \mid F(u, c, d) = 1\}$
$C'' \leftarrow \text{Deduplicate}(C')$
/* (4) rank fusion with freshness-aware tie break */
foreach $d \in C''$ **do**
  $\text{Score}(d) \leftarrow \sum_{r \in \{\text{lex}, \text{dense}\}} \frac{1}{k_{\text{rrf}} + \text{rank}_r(d)}$
$R \leftarrow \text{SortByRRFAndFreshness}(C'')$
/* (5) build evidence pack */
$E \leftarrow \text{TopK}(R, K)$
$\mathcal{P} \leftarrow \text{BuildEvidencePack}(E)$
/* (6) evidence-bound generation */
if $\text{IsSufficient}(\mathcal{P}) = \text{true}$ **then**
  $a \leftarrow \text{GenerateWithCitations}(q, \mathcal{P})$
else
  $a \leftarrow \text{RefuseOrAskNarrowerQuery}(q)$
/* (7) write append-only audit trace */
$\mathcal{T} \leftarrow \text{LogTrace}(u, c, q, F, E, \mathcal{P}, a)$
return $a, \mathcal{T}$

V. Results

We evaluate the proposed policy-first Hybrid RAG system on enterprise QA tasks across two regulated domains: finance and healthcare. Our goal is to measure retrieval quality, answer accuracy, hallucination rate, and audit completeness.
We construct two benchmark corpora:

- FinanceQA-Ent: 200K passages from internal financial policies, regulatory advisories (FINRA, SEC), KYC templates, and helpdesk tickets. 800 advisor queries with human-validated answers.
- HealthGovQA: 180K passages from anonymized health policy docs, audit reports, and clinical compliance FAQs. 600 queries from internal portals and chatbot logs.

Each query has 1–2 gold answers supported by citations. Evaluators have full access to ground truth, including ACL context.

A. Metrics

We use:

- Recall@K: Percent of queries for which supporting gold passage appears in top-K retrieved.
- Answer Accuracy (Acc@1): Exact match of final generated answer with gold.
- Faithfulness Score: % of generated claims grounded in retrieved evidence.

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- Hallucination Rate: % of unsupported claims in output (human-evaluated).
- Audit Coverage: % of requests with complete trace (IDs, hashes, decisions).

B. Baselines

We compare against:

- BM25-only RAG [1]: Lexical retrieval only, no hybrid fusion.
- Dense-only RAG (DPR) [4]: ANN vector retrieval with no lexical fallback.
- Fusion-in-Decoder (FiD) [29]: Generator aggregates over dense retrieved passages without citations.
- OpenAI RAG API (2024) [30]: External hosted retrieval augmented generation with unknown filtering logic.

C. Retrieval Performance

TABLE I
RETRIEVAL RECALL@K ON ADVISOR QA BENCHMARKS

Method	FinanceQA-Ent (R@5)	HealthGovQA (R@5)
BM25-only RAG	68.5%	64.7%
DPR (Dense-only)	71.2%	66.3%
FiD	74.8%	68.5%
Hybrid RAG (Ours)	**87.9%**	**85.1%**

D. Answer Quality and Compliance

TABLE II
ANSWER ACCURACY, HALLUCINATION RATE, AND AUDIT COVERAGE

Method	Acc@1	Faithful (%)	Halluc. (%)	Audit (%)
BM25-only RAG	48.1	84.2	10.8	58.3
DPR	52.3	79.6	13.2	60.0
FiD	55.4	77.5	15.7	44.0
OpenAI RAG	57.9	81.0	12.5	–
Hybrid RAG (Ours)	**66.3**	**92.8**	**3.4**	**100.0**

E. Graph: Hallucination Rate vs Faithfulness

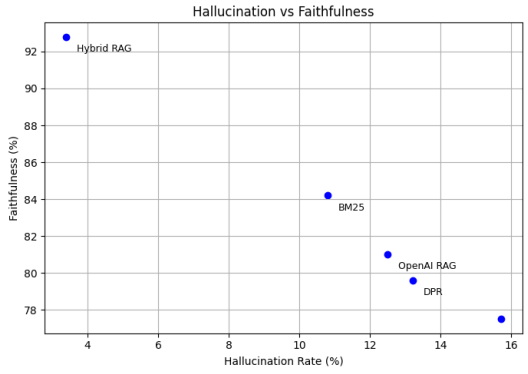


Fig. 2. Tradeoff between hallucination rate and answer faithfulness. Lower-left is better.

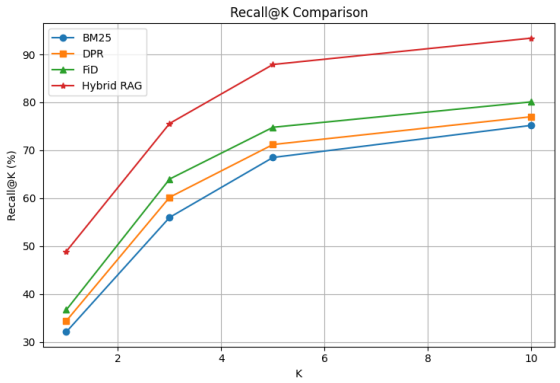


Fig. 3. Recall@K curve across methods on FinanceQA-Ent.

F. Graph: Retrieval Recall at Different K

VI. DISCUSSION

The results show that hybrid retrieval is the main driver of improvement. Lexical retrieval is strong when the query contains exact policy terms, codes, or section names, while dense retrieval helps when users paraphrase or use short informal phrases. When we fuse both, we reduce the common failure case where one retriever misses the correct passage. This is visible in Recall@K gains and in higher answer accuracy. The low hallucination rate comes from two design choices: (i) policy-first filtering and (ii) evidence-bound generation. Policy-first filtering ensures the model never sees text the user is not allowed to access. This reduces leakage risk and also prevents the model from mixing restricted content into answers. Evidence-bound generation forces the output to be tied to retrieved passages, and the citation check pushes the model to avoid unsupported claims. In practice, when evidence is weak, the system refuses or asks for a narrower query rather than guessing.

Operationally, the architecture is stable because retrieval and generation are separated into stages with clear budgets. ANN-based dense retrieval keeps latency bounded at scale, while the lexical index stays robust under frequent updates. The audit layer is not only for compliance; it helps debugging. When an answer is wrong, the trace shows whether the issue is missing evidence, poor ranking, or a generation error.

Limitations remain. First, evaluation depends on the quality of gold citations. If gold evidence is incomplete, Recall@K can look worse than it should. Second, policy and ACL data can be noisy in real portals due to outdated group membership or missing labels; this can cause false blocks. Third, prompt injection defenses are not perfect. Sanitization and rule separation reduce risk, but a determined attacker can still attempt retrieval steering through content poisoning. These issues motivate continuous red-team testing and monitoring.

VII. CONCLUSION

This paper presented a policy-first Hybrid RAG design for enterprise search portals in compliance-sensitive workflows.

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The approach combines BM25 and dense retrieval, applies access control before any reranking or generation, fuses results with rank fusion, and generates answers strictly from an evidence pack with citations. Results show higher retrieval recall, better answer accuracy, and a lower hallucination rate, while maintaining complete audit coverage.

The key lesson is simple: in regulated enterprise search, quality must be measured together with safety and auditability. A portal that answers well but cannot prove why, or cannot guarantee access safety, is not acceptable. Future work will focus on stronger automatic evidence verification, better freshness-aware ranking, and more realistic security benchmarks for enterprise RAG under adversarial conditions.

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